



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2024-25(Even Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

End Semester

Semester: VI

Programme: B. Tech Information Technology

Course: Ethical Hacking

Time: 3 hour

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33603	Course Outcome
CO1	To study the fundamental concepts of ethical hacking, including various phases and types of hackers
CO2	To identify threats, vulnerabilities, and attacks in computer systems and networks
CO3	To explore tools and techniques used in ethical hacking such as foot printing, scanning, social engineering, and password cracking
CO4	To evaluate system and network security measures using penetration testing and vulnerability assessment tools
CO5	Demonstrate awareness of the legal, ethical, and professional issues related to cybersecurity practices

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define footprinting and give one technique.	CO1	3	2
b.	What is configuration flaw?	CO2	2	2
c.	Define SQL Injection	CO3	2	2
d.	What is cyber law?	CO4	3	2
e.	What is steganography ?	CO5	2	2

Group B

Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain phases of ethical hacking with neat diagram.	CO1	4	5
b.	Compare OSI and TCP/IP models with advantages.	CO1	5	5
c.	Explain social engineering attacks with real-life examples.	CO1	4	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Analyze different types of vulnerabilities in a system with examples.	CO2	4	5
b.	Explain wireless network security challenges and solutions.	CO2	5	5
c.	Differentiate between malware-based and network-based attacks.	CO2	4	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain penetration testing methodology in detail.	CO3	4	5
b.	Describe SQL Injection and XSS with prevention techniques.	CO3	5	5
c.	Explain IDS, IPS, and firewall with working differences.	CO3	4	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Discuss legal and ethical issues in ethical hacking.	CO2	4	5
b.	Explain professional integrity in cybersecurity with examples	CO2	5	5
c.	Analyze misuse of hacking tools and its consequences.	CO2	4	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain password cracking techniques in detail.	CO2	4	5
b.	Describe DoS attack and its types with prevention.	CO2	5	5
c.	Explain cryptography techniques and their importance.	CO2	4	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create